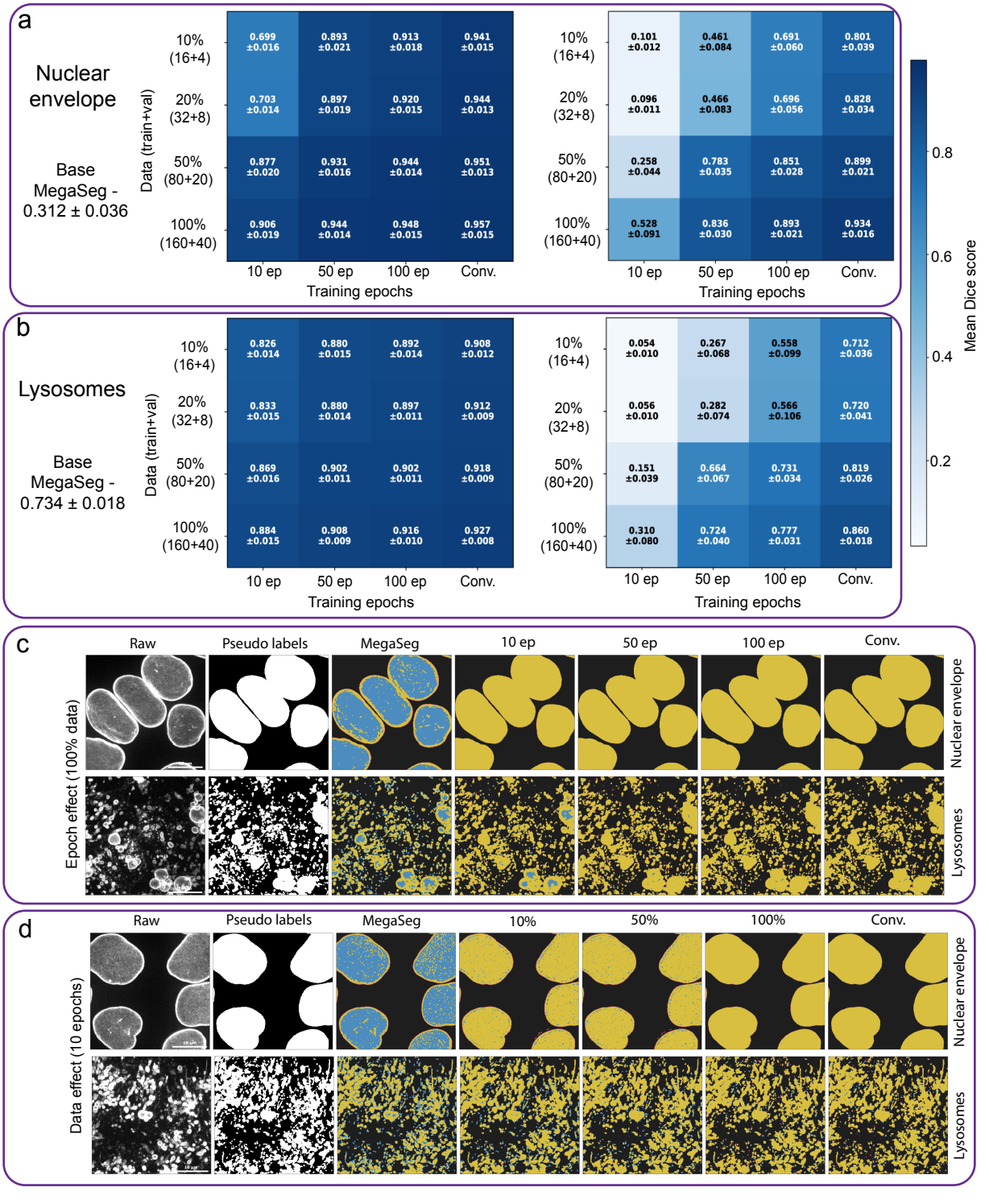

a
Nuclear envelope
Base MegaSeg - 0.312 ± 0.036
Data (train+val)
10% (16+4)
20% (32+8)
50% (80+20)
100% (160+40)
0.699 ±0.016
0.893 ±0.021
0.913 ±0.018
0.941 ±0.015
0.703 ±0.014
0.897 ±0.019
0.920 ±0.015
0.944 ±0.013
0.877 ±0.020
0.931 ±0.016
0.944 ±0.014
0.951 ±0.013
0.906 ±0.019
0.944 ±0.014
0.948 ±0.015
0.957 ±0.015
10 ep
50 ep
100 ep
Conv.
Training epochs
0.101 ±0.012
0.461 ±0.084
0.691 ±0.060
0.801 ±0.039
0.096 ±0.011
0.466 ±0.083
0.696 ±0.056
0.828 ±0.034
0.258 ±0.044
0.783 ±0.035
0.851 ±0.028
0.899 ±0.021
0.528 ±0.091
0.836 ±0.030
0.893 ±0.021
0.934 ±0.016
Mean Dice score
0.8
0.6
0.4
0.2
b
Lysosomes
Base MegaSeg - 0.734 ± 0.018
0.826 ±0.014
0.880 ±0.015
0.892 ±0.014
0.908 ±0.012
0.833 ±0.015
0.880 ±0.014
0.897 ±0.011
0.912 ±0.009
0.869 ±0.016
0.902 ±0.011
0.902 ±0.011
0.918 ±0.009
0.884 ±0.015
0.908 ±0.009
0.916 ±0.010
0.927 ±0.008
0.054 ±0.010
0.267 ±0.068
0.558 ±0.099
0.712 ±0.036
0.056 ±0.010
0.282 ±0.074
0.566 ±0.106
0.720 ±0.041
0.151 ±0.039
0.664 ±0.067
0.731 ±0.034
0.819 ±0.026
0.310 ±0.080
0.724 ±0.040
0.777 ±0.031
0.860 ±0.018
c
Raw
Pseudo labels
MegaSeg
10 ep
50 ep
100 ep
Conv.
Epoch effect (100% data)
Nuclear envelope
Lysosomes
d
Raw
Pseudo labels
MegaSeg
10%
50%
100%
Conv.
Data effect (10 epochs)
Nuclear envelope
Lysosomes